


 रोल नं.

--	--	--	--	--	--	--	--

 Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

नोट
 (I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित (I) पृष्ठ **23** हैं।

 (II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में (II) **33** प्रश्न हैं।
NOTE
 Please check that this question paper contains **23** printed pages.

 Please check that this question paper contains **33** questions.

❖ (III) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए (III) प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।

❖ (IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से (IV) पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।

❖ (V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का (V) समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।

Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.

Please write down the serial number of the question in the answer-book before attempting it.

15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

रसायन विज्ञान (सैद्धांतिक)

(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

CHEMISTRY (Theory)

(FOR VISUALLY IMPAIRED CANDIDATES ONLY)

निर्धारित समय : 3 घण्टे

Time allowed : 3 hours

अधिकतम अंक : 70

Maximum Marks : 70

General Instructions :

Read the following instructions carefully and follow them :

- (i) *This question paper contains **33** questions. **All** questions are **compulsory**.*
- (ii) *This question paper is divided into **five** sections – **Sections A, B, C, D and E**.*
- (iii) *In **Section A** – Questions no. **1** to **16** are Multiple Choice type questions. Each question carries **1** mark.*
- (iv) *In **Section B** – Questions no. **17** to **21** are Very Short Answer type questions. Each question carries **2** marks.*
- (v) *In **Section C** – Questions no. **22** to **28** are Short Answer type questions. Each question carries **3** marks.*
- (vi) *In **Section D** – Questions no. **29** and **30** are case study-based questions. Each question carries **4** marks.*
- (vii) *In **Section E** – Questions no. **31** to **33** are Long Answer type questions. Each question carries **5** marks.*
- (viii) *There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the Sections except Section A.*
- (ix) *Use of calculators is **not** allowed.*

Section A

*Questions no. **1** to **16** are Multiple Choice type Questions, carrying **1** mark each.*

$$16 \times 1 = 16$$

1. Out of the following, the strongest base in aqueous solution is :

- (A) Methylamine
- (B) Dimethylamine
- (C) Trimethylamine
- (D) Aniline

2. Iodoform test is given by
- (A) Propanol (B) Ethyl chloride
(C) Pentanal (D) Pentan-2-one
3. Out of the following transition elements, the maximum number of oxidation states is shown by
- (A) Sc (B) Cr
(C) Mn (D) Fe
4. Which of the following cell was used in Apollo space programme ?
- (A) Dry cell (B) Ni – Cd cell
(C) $H_2 - O_2$ fuel cell (D) Mercury cell
5. Consider the following standard electrode potential values :
- $$Fe^{3+} + 1e^{-} \longrightarrow Fe^{2+}; E^{\circ} = + 0.77 \text{ V}$$
- $$MnO_4^{-} + 8H^{+} + 5e^{-} \longrightarrow Mn^{2+} + 4H_2O; E^{\circ} = + 1.51 \text{ V}$$
- What is the cell potential for the redox reaction ?
- (A) + 2.28 V (B) – 0.74 V
(C) – 2.28 V (D) + 0.74 V
6. Which of the following statements is **not** true about glucose ?
- (A) It is an aldohexose.
(B) It gives 2,4-DNP test.
(C) It is present in pyranose form.
(D) On heating with HI it forms n-hexane.

7. Amides can be converted into amines by the reaction named
- (A) Diazotisation
 - (B) Carbylamine
 - (C) Ammonolysis
 - (D) Hoffmann Bromamide degradation
8. Which of the following would be a better choice for reducing nitrobenzene to aniline ?
- (A) H_2/Ni
 - (B) LiAlH_4
 - (C) Fe and HCl
 - (D) Sn and HCl
9. Which property of transition metals enables them to behave as catalysts ?
- (A) Variable oxidation states
 - (B) Alloy formation
 - (C) High ionisation enthalpy
 - (D) High melting point
10. Which of the following is the most stable complex species ?
- (A) $[\text{Fe}(\text{CN})_6]^{3-}$
 - (B) $[\text{Fe}(\text{CO})_5]$
 - (C) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$
 - (D) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$

11. The role of a catalyst is to change :

- (A) equilibrium constant (B) $\Delta_r G$
(C) $\Delta_r H$ (D) E_a

12. Van't Hoff factor values for KCl and K_2SO_4 respectively are :

- (A) 1 and 2 (B) 2 and 3
(C) 2 and 2 (D) 2 and 4

For Questions number 13 to 16, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

13. *Assertion (A)* : For a zero order reaction, the unit of rate constant and rate of reaction are same.

Reason (R) : Rate of reaction for zero order reaction is independent of concentration of reactant.

14. *Assertion (A)* : o-nitrophenol is a weaker acid than p-nitrophenol.

Reason (R) : Intramolecular hydrogen bonding makes para-isomer stronger acid than ortho-isomer.

15. *Assertion (A)* : $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$ gives a white precipitate with silver nitrate solution.

Reason (R) : The complex dissociates to give Cl^- and SO_4^{2-} ions.

16. *Assertion (A)* : Chloroform and acetone mixture shows negative deviation from Raoult's law.

Reason (R) : In chloroform and acetone mixture, A – A or B – B type intermolecular interactions are weaker than A – B type interactions.

Section B

17. State Henry's law and write its two applications. 2

18. (a) Define the following terms : $2 \times 1 = 2$

(i) Crystal field splitting energy

(ii) Ambidentate ligand

OR

(b) What is meant by the chelate effect ? Give an example. 2

19. Name the reagents used in the following reactions : $2 \times 1 = 2$

(a) Bromination of phenol to 2,4,6-tribromophenol

(b) Conversion of Butan-2-one to Butan-2-ol

20. Predict the products formed when propanone reacts with following reagents : $2 \times 1 = 2$
- (a) CH_3MgBr and then H_3O^+
- (b) Zinc amalgam and concentrated HCl
21. What happens when D-glucose is treated with the following reagents : $2 \times 1 = 2$
- (a) HCN
- (b) Br_2 water

Section C

22. An antifreeze solution is prepared by dissolving 31 g of ethylene glycol (Molar mass = 62 g mol^{-1}) in 600 g of water. Calculate the freezing point of the solution. (K_f for water = $1.86 \text{ K kg mol}^{-1}$) 3
23. Conductivity of $2 \times 10^{-3} \text{ M}$ methanoic acid is $8 \times 10^{-5} \text{ S cm}^{-1}$. Calculate its molar conductivity and degree of dissociation if Λ_m^0 for methanoic acid is $404 \text{ S cm}^2 \text{ mol}^{-1}$. 3
24. (a) Give the IUPAC name and electronic configuration of central metal atom in terms of t_{2g} and e_g of $\text{K}_4[\text{Mn}(\text{CN})_6]$.
- (b) How many ions are produced from the complex $[\text{Cr}(\text{NH}_3)_6] \text{Cl}_2$ in solution ? $2 + 1 = 3$

25. Justify and arrange the following compounds of each set in increasing order of reactivity towards the asked displacement :
- (a) 1-Bromobutane, 2-Bromobutane, 2-Bromo-2-Methylpropane
(S_N1 reaction)
- (b) 2-Bromo-2-Methylbutane, 1-Bromopentane, 2-Bromopentane
(S_N2 reaction) $2 \times 1 \frac{1}{2} = 3$
26. Define the following terms : $3 \times 1 = 3$
- (a) Racemic mixture
- (b) Chiral carbon
- (c) Enantiomers
27. Explain the following : $3 \times 1 = 3$
- (a) Reimer-Tiemann reaction
- (b) Kolbe's reaction
- (c) Williamson synthesis
28. Account for the following : (Any *three*) $3 \times 1 = 3$
- (a) pK_b of aniline is more than that of methylamine.
- (b) Ethylamine is soluble in water whereas aniline is not.
- (c) Aniline does not undergo Friedel-Crafts reaction.
- (d) Aromatic primary amines cannot be prepared by Gabriel phthalimide synthesis.

Section D

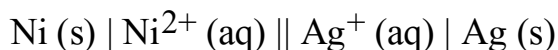
Questions number 29 and 30 are case study-based questions. Read the following paragraphs and answer the questions that follow.

- 29.** Organic compounds containing amine as functional group are present in a vivid variety of compounds, namely amino acids, hormones, neurotransmitters, DNA, dyes, etc. Amino acids are classified as acidic, basic or neutral depending upon the relative number of amino and carboxyl groups in their molecule. In aqueous solution, the carboxyl group can lose a proton and amino group can accept a proton giving rise to a dipolar ion known as zwitter ion. In zwitter ionic form, amino acids show amphoteric behaviour.
- (a) Give one point of difference between acidic and basic amino acid. 1
- (b) Name the linkage formed when carboxyl end of one amino acid condenses with amino end of other amino acid. 1
- (c) (i) What are essential amino acids and non-essential amino acids ? 2
- OR**
- (c) (ii) What is meant by zwitter ionic form of amino acids ? Why do amino acids show amphoteric behaviour ? 2
- 30.** Oxidation-reduction reactions are commonly known as redox reactions. In galvanic cell, the chemical energy of a spontaneous redox reaction is converted into electrical energy, whereas in an electrolytic cell, electrical energy is used to carry out a non-spontaneous redox reaction. The standard electrode potential for any electrode dipping in an appropriate solution is defined with respect to standard electrode potential of hydrogen electrode taken as zero. Concentration dependence of the potentials of the electrodes and the cells are given by Nernst equation.

- (a) Predict whether the reaction is spontaneous or non-spontaneous if E_{cell}° is positive ? 1
- (b) When does a galvanic cell behave like an electrolytic cell ? 1
- (c) (i) Define the following terms : 2×1=2
- (1) Redox reaction
- (2) Galvanic cell

OR

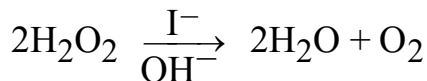
- (c) (ii) Write cell reaction and Nernst equation for the following cell at 25°C : 2



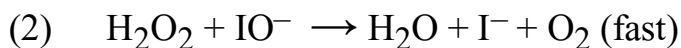
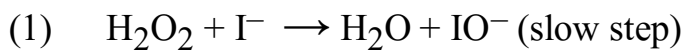
Section E

31. (a) (i) What is rate of reaction ? Write two factors that affect the rate of reaction.

- (ii) For a reaction,



the proposed mechanism is as given below :



(I) Write rate law for the reaction.

(II) Write the overall order of reaction.

(III) Out of steps (1) and (2), which one is rate determining step ? 2+3=5

OR

- (b) (i) Define pseudo first order reaction with one example.
- (ii) The rate constant of a first order reaction increases from 4×10^{-2} to 8×10^{-2} when the temperature changes from 27°C to 37°C . Calculate its energy of activation (E_a). $2+3=5$
- (Given : $2.303 R = 19.15 \text{ JK}^{-1} \text{ mol}^{-1}$, $\log 2 = 0.30$)

32. Attempt any *five* of the following : $5 \times 1 = 5$

- (a) Out of Cr^{3+} and Mn^{3+} , which one is a strong oxidising agent and why ?
- (b) Which ion amongst the following is colourless and why ?
 Ti^{4+} , Cr^{3+} , V^{3+}
(Atomic number : Ti = 22, Cr = 24, V = 23)
- (c) Why is Ce^{4+} a strong oxidising agent ?
- (d) Why is highest oxidation state of a metal shown in oxide or fluoride only ?
- (e) Write one similarity between lanthanoids and actinoids.
- (f) Explain why Cu^+ ion is not stable in aqueous solution.
- (g) Why do transition elements have high enthalpies of atomisation ?

33. (a) Write short notes on the following :

5×1=5

- (i) Wolff-Kishner Reduction
- (ii) Aldol Condensation
- (iii) Rosenmund Reduction
- (iv) Cannizzaro Reaction
- (v) Hell-Volhard-Zelinsky Reaction

OR

(b) (i) What happens when

- (1) CH_3COCl is treated with $(\text{CH}_3)_2\text{Cd}$,
- (2) CH_3CHO is treated with HCN , and
- (3) benzaldehyde is treated with conc. $(\text{HNO}_3 + \text{H}_2\text{SO}_4)$?

(ii) Which acid of each pair shown below would you expect to be stronger ?

- (1) Acetic acid or fluoroacetic acid
- (2) p-nitrobenzoic acid or p-methylbenzoic acid

3+2=5